

Lorenzo Veronese

Computer Engineer & Bioengineer

Como, Italy
✉ lorenzo.veronese@polimi.it
🌐 lorenzoveronese.github.io
in [lorenzo-v-veronese](https://www.linkedin.com/in/lorenzo-v-veronese)
ORCID: 0000-0001-9204-3363

Profile

PhD Candidate at Politecnico di Milano, specializing in Computational Cognitive Neuroscience and Neuro-inspired AI. My research focuses on developing computational models to understand human cognitive functions such as planning (through theory-driven models of hippocampal-prefrontal interactions) and visual perception (via generative neural decoding architectures). Recent publications include enhancements to fMRI-based neural decoding pipelines (more than 95% improvement in computational efficiency) and development of ROI-centric explainability analyses.

Education

- 2024–Present **PhD, Bioengineering**, *Politecnico di Milano (Milan, Italy)*, Supervisors: Prof. Pietro Cerveri, Dr. Pasquale Anthony Della Rosa, Computational Cognitive Neuroscience for Understanding Human Planning and Perception.
- 2022–2024 **M.Sc., Computer Science Engineering, Artificial Intelligence**, *Politecnico di Milano (Milan, Italy)*, Thesis: "Visual stimulus reconstruction from BOLD fMRI signal using generative AI", Final grade: 110 *cum laude*/110 (GPA: 29.2).
- 2019–2022 **B.Sc., Computer Science Engineering**, *Politecnico di Milano (Milan, Italy)*, Final grade: 103/110.

Research Experience

- 2024–Present **PhD Candidate**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering)*, collaborating with *IRCCS Ospedale San Raffaele (Department of Neuroradiology)*.
- Computational cognitive neuroscience: Designing a theory-driven neural model of human planning in the Tower of London neuropsychological task.
 - Neuro-AI generative modeling: Designing an explainable two-stage generative decoding pipeline to reconstruct visual perception from fMRI data.
- 2021–2023 **Student Researcher**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering)*, *NECSTLab*.
- Generative AI for data harmonization: Engineered a CycleGAN-based deep learning framework for the harmonization of histopathological images across microscopes.
- 2022–2023 **Student Researcher**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering)*, *DEEP-SE Lab*.
- Human-computer interaction in psychotherapy: Modeled the dynamics of psychotherapy conversations using formal modelling and natural language processing.
- 2021–2022 **Student Researcher**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering)*, *AIRLab*.

- Robotic design and prototyping: Led the mechanical and software design of an assistive robotic device to support individuals with motor disabilities in writing tasks.

Scientific Publications

- Journal **Veronese, L.**, Moglia, A., Pecco, N., Della Rosa, P.A., Scifo, P., Mainardi, L., & Cerveri, P. (2025). Optimized AI-based neural decoding from BOLD fMRI signal for analyzing visual and semantic ROIs in the human visual system. *Journal of Neural Engineering*, 22(4). 10.1088/1741-2552/adfbc2.
- Della Rosa, P.A., Pecco, N., Calloni, S.F., Canini, M., Oprandi, C., Pavani, D., Grimaldi, G., **Veronese, L.**, Favalli, V., Frontino, G., Gambarini, A., Olivieri, G., Scotti, R., Pontesilli, S., Ogliari, A., Rigamonti, A., Falini, A., Baldoli, C., & Bonfanti, R. (2026). Multimodal MRI Neurodevelopmental Profiling in Type 1 Diabetes: Long-term Effects of MDI vs CSII Treatments. *The Journal of Clinical Endocrinology & Metabolism*. dgag102.
- Conference **Veronese, L.**, Pecco, N., Moglia, A., Scifo, P., Castellano, A., Mainardi, L., Della Rosa, P.A., & Cerveri, P. (2026). Visual neural decoding performance on different semantic classes of stimuli. *Associazione Italiana Risonanza Magnetica in Medicina (AIRMM 2026)*. (Oral presentation)
- Veronese, L.**, Pecco, N., Moglia, A., Scifo, P., Castellano, A., Mainardi, L., Della Rosa, P.A., & Cerveri, P. (2025). Evaluating Semantic Brain Regions Contribution to Visual Neural Decoding Performance on Different Classes of Stimuli. *20th conference on Computational Intelligence methods for Bioinformatics and Biostatistics (CIBB 2025)*. (Oral presentation. Extended version in press on Lecture Notes in Bioinformatics)
- Veronese, L.**, Poles, I., D'Arnese, E., Santambrogio, M.D. (2023). Stain Transfer using CycleGAN for Histopathological Images. *IEEE EUROCON 2023-20th International Conference on Smart Technologies*. 10.1109/EUROCON56442.2023.10199027.
- Preprint Dettori, F.*, Forasassi, M.*, **Veronese, L.***, Lestingi, L., Scotti, V., & Rossi, M.G. (2026). Do You Understand How I Feel?: Towards Verified Empathy in Therapy Chatbots. *ArXiv*. 10.48550/arXiv.2601.08477. (*Equal contribution)

Teaching

- 2025 **Teaching Assistant, Neuroengineering, Politecnico di Milano**
- Delivered lectures on deep learning architectures and methodologies.
 - Supervised student groups in designing and training end-to-end deep learning models for decoding visual categories from fMRI data.
- 2025 **Teaching Assistant, Fundamentals of Computer Science, Politecnico di Milano**
- Led laboratory sessions for undergraduate students, teaching algorithmic thinking and data structures in C programming.

Peer Reviewing

- Journal IEEE Transactions on Image Processing; IEEE Transactions on Fuzzy Systems; IEEE Transactions on Cognitive and Developmental Systems.
- Conference Cognitive Computational Neuroscience (CCN 2026).

Awards & Honors

- 2025 Selected by Nova Talent (top 3% talent network).
- 2022 Scholarship for academic merit — Town of Cucciago & Rubelli S.p.a.

Advanced Training & Summer Schools

- 2025 **European Course on Advanced Imaging Techniques in Neuroradiology**, *IRCCS Ospedale San Raffaele*, Milan, Italy
- 2025 **XLIV Annual School of Bioengineering**, *University of Pisa*, Pisa, Italy
Topic: "Unlocking the Mind and Emotions: Cutting-Edge Bioengineering Techniques from Computational Physiology to Clinical Applications"

Skills

- Expertise Computational Cognitive Neuroscience, Artificial Intelligence, Deep Learning, fMRI, Planning
- Programming Python, PyTorch, TensorFlow, MATLAB, Bash
- Tools Git, LaTeX, SPM, FSL
- Languages English (C1; TOEFL iBT score: 5.5/6), Italian (native)

References

- Pietro Cerveri Full Professor — Politecnico di Milano & Università di Pavia
pietro.cerveri@unipv.it
- Pasquale A. Della Rosa Leading Researcher — IRCCS Ospedale San Raffaele
dellarosa.pasquale@hsr.it

I hereby authorize the processing of my personal data included in this CV pursuant to Art. 13 of Legislative Decree 196/2003 and Art. 13 of GDPR (EU Regulation 2016/679).